

Problem-Solution Fit

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Individuals applying for credit cards. • Banks and financial institutions. • Loan and credit verification officers.	6. CUSTOMER LIMITATIONS [CL] • Limited knowledge of bank eligibility criteria. • Different income and employment backgrounds. • Limited access to financial guidance. • Time constraints during the application process.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Verification • Pros: Human judgment and detailed verification. • Cons: Slow, time-consuming, and prone to human errors. Traditional Rule-Based Systems • Pros: Easy to implement. • Cons: Less accurate and cannot learn from historical data.
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2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Customers do not know whether they are eligible for a credit card.
- Banks receive a large number of applications every day.
- Manual verification takes more time and may lead to inconsistent decisions

9. PROBLEM ROOT / CAUSE [RC]

- Complex eligibility criteria.
- Large number of applications.
- Manual verification process.
- Lack of automated prediction systems.
- Risk of approving ineligible applicants.

7. BEHAVIOR + ITS INTENSITY [BE]

- Customers compare eligibility before applying.
- Banks verify every application carefully.
- High demand for quick and accurate approval decisions.

3. TRIGGERS TO ACT [TR]

- Customer wants to apply for a new credit card.
- Bank needs a faster screening process.
- Need to reduce manual work and improve decision-making.

10. YOUR SOLUTION [SL]

- Develop a **Credit Card Approval Prediction System** using Machine Learning.
- Preprocess applicant data and perform feature engineering.
- Train multiple machine learning models and select the best-performing model (Random Forest).
- Predict whether a credit card application is **Approved** or **Rejected**.
- Display the prediction along with a confidence score through a Flask web application.
- Help banks make faster and more consistent approval decisions while helping customers understand their eligibility.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Bank websites
- Flask web application
- Internet banking portals

OFFLINE

- Bank branches
- Customer service centers

4. EMOTIONS [EM]

Before

- Confused
- Anxious
- Uncertain

After

- Confident
- Satisfied
- Informed about eligibility